GigaWorld-Policy: An Efficient Action-Centered World--Action Model

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https://arxiv.org/abs/2603.17240

Research Questions

- How can World-Action Models (WAMs) be made more inference-efficient while maintaining or improving policy learning performance?
- How can future visual dynamics supervision be leveraged without requiring expensive video generation at inference time?
- What training paradigm can effectively adapt video generation models for action-centered robot policy learning?

Core Idea

Existing World-Action Models jointly predict actions and future videos, but this creates two problems: expensive inference requiring video generation at every step, and action prediction being dependent on potentially noisy video forecasts. GigaWorld-Policy solves this with a clever causal masking design that trains the model to predict both actions and future videos jointly (providing dense supervision), but prevents future video tokens from influencing action tokens. This allows the model to skip video generation entirely at inference time while still benefiting from the richer training signal during learning.

Key Formula

Joint Training Loss

$$\mathcal{L}_{all} = \lambda_{video} \mathcal{L}_{video} + \lambda_{action} \mathcal{L}_{action}$$

Combined loss function where λ_video and λ_action are scalar weights balancing video prediction loss $\mathcal{L}$_video and action prediction loss $\mathcal{L}$_action. This enables joint supervision from both modalities during training while allowing action-only inference.

Action Flow Integration

$$\frac{d a^{(s)}}{d s} = g_{\Theta}\left(a^{(s)}, s \mid w_t\right), s \in [0, 1]$$

Flow matching integration for action generation where a^(s) is the action at flow time s, g_Θ is the learned velocity field, and w_t is the context (language, state, observation). This generates actions without requiring video token instantiation.

Key Figure

Figure 2: (a) VLM-based VLA with auxiliary future supervision to densify training signals, but limited by the discriminative nature of VLMs (Cen et al., 2025; Ni et al., 2025; Zhang et al., 2025). (b) Joint action–video prediction with bidirectional attention (Bi et al., 2025; Kim et al., 2026; Shen et al., 2025), requiring future video generation at inference. (c) Two-stage design (Pai et al., 2025) that first generates future video and then gets actions via an Inverse Dynamics Model (IDM), inheriting video prediction errors and incurring additional inference cost. (d) GigaWorld-Policy: an action-centered WAM that leverages future visual dynamics as supervision during training, while making future-video prediction optional at inference for low-latency action generation.

action sequences from the current observation context, and in parallel learns an action-conditioned visual dynamics model that forecasts future visual observations given the same current observation together with the predicted actions, thereby coupling action learning with explicit 2D pixel-level state evolution. These two learning signals are optimized within the same model, allowing future visual dynamics to regularize action plausibility and provide substantially denser supervision, which improves learning efficiency. Crucially, at inference time, explicit future-video prediction is optional: the model can be executed in an action-only mode that directly produces control commands without rolling out long sequences of video tokens. This design substantially reduces compute and memory overhead, avoids compounding errors from extended visual rollouts, and enables low-latency closed-loop control, as illustrated in Fig. 2 (d). To obtain stronger pre-trained weights, we use a curriculum training pipeline that injects physics priors from diverse video sources before any task-specific supervision. GigaWorld-Policy is initialized from a large-scale web-video foundation model (Wan et al., 2025), and then further pre-trained on embodied, robot-centric data that combines real robot recordings with large-scale egocentric human videos, improving robustness to embodiment-specific viewpoints and interaction dynamics. Finally, we post-train the model on target-robot trajectories that align images, language, and actions, specializing it for instruction-conditioned action prediction under the target robot's control interface and state distribution.

We validate GigaWorld-Policy through experiments in both simulation and real-world environments. GigaWorld-Policy outperforms strong baselines, delivering efficiency gains without sacrificing control performance. As shown in Fig. 1, it achieves a trade-off between success and efficiency, delivering $9\times$ faster inference and 7% higher task success rates than the state-of-the-art WAM Motus (Bi et al., 2025). Moreover, under comparable speed and VLA settings, GigaWorld-Policy improves performance by 20%.

The main contributions of this paper are summarized as follows:

- We propose GigaWorld-Policy, an action-centered and efficient World–Action model. During training, future visual dynamics provide dense supervision and a reasoning signal for action learning, without over-reliance on explicit video synthesis. At inference time, future-visual prediction is optional and we can decode actions only, enabling low-latency control.
- We propose a pre-training paradigm that converts a generic video generation model into a strong initialization for robot policy learning, fully leveraging complementary data sources across stages.
- Experiments on real robotic platforms show that GigaWorld-Policy achieves a $9\times$ inference speedup (down to $0.36\,\mathrm{s}$ per inference) while improving task success rates by 7% over baseline methods; compared with $\pi_{0.5}$, GigaWorld-Policy matches the inference speed and improves performance by 95% on RoboTwin 2.0.

Figure 2: Shows the key architectural difference: GigaWorld-Policy uses causal masking to make future video prediction optional at inference, unlike existing WAMs that require video generation.

Limitations

- Requires careful tuning of loss weights λ_video and λ_action to balance supervision signals effectively
- Still depends on accurate current observation encoding - errors in visual perception can propagate to action prediction

- Causal masking design may limit the model's ability to leverage full bidirectional reasoning between action and video modalities
- Performance gains may diminish for very long-horizon tasks where video prediction errors would compound anyway